

openGUTS Report

Project:

GUTS_RF_Hyalella_45_Noprefitbackgroundhazard_29052026

Project file:

No project file saved or loaded

Project description (optional):

No project description available

Software version:

openGUTS - 1.2

Date of report creation:

01/06/2026 11:06:31

Calibration

Calibration input data

Data set 1

File: Data edited manually

Description (optional):

Hyalella

Control group: 'Control'

Survival data of input data set 1:

Time [d]	Control	T1	T2	T3	T4	T5	T6
0	80	20	20	29	20	30	20
0.042	80	20	20	29	20	30	11
0.083	80	20	20	29	19	18	4
0.16	80	20	20	29	19	14	0
0.33	79	20	20	29	19	5	0
1	79	20	20	29	19	0	0
1.208	79	20	20	29	18	0	0
2	78	20	20	29	16	0	0
3	78	20	20	28	14	0	0
4	78	20	20	28	14	0	0

Concentration data of input data set 1:

Time [d]	Control	T1	T2	T3	T4	T5	T6
0	0	0.3	1	3	10	30	100
0.2	0	0.3	1	2.9	9.6	28.8	96.1
0.4	0	0.3	0.9	2.8	9.2	27.7	92.3
0.6	0	0.3	0.9	2.7	8.9	26.6	88.7
0.8	0	0.3	0.9	2.6	8.5	25.6	85.2
1	0	0.2	0.8	2.5	8.2	24.6	81.9
1.2	0	0.2	0.8	2.4	7.9	23.6	78.7
1.4	0	0.2	0.8	2.3	7.6	22.7	75.6
1.6	0	0.2	0.7	2.2	7.3	21.8	72.7

1.8	0	0.2	0.7	2.1	7	20.9	69.8
2	0	0.2	0.7	2	6.7	20.1	67.1
2.2	0	0.2	0.6	1.9	6.4	19.3	64.5
2.4	0	0.2	0.6	1.9	6.2	18.6	61.9
2.6	0	0.2	0.6	1.8	6	17.9	59.5
2.8	0	0.2	0.6	1.7	5.7	17.2	57.2
3	0	0.2	0.5	1.6	5.5	16.5	54.9
3.2	0	0.2	0.5	1.6	5.3	15.8	52.8
3.4	0	0.2	0.5	1.5	5.1	15.2	50.7
3.6	0	0.1	0.5	1.5	4.9	14.6	48.7
3.8	0	0.1	0.5	1.4	4.7	14	46.8
4	0	0.1	0.4	1.3	4.5	13.5	45

Calibration settings

Calibration parameter settings for GUTS-RED-SD:

Parameter	Fit	Min	Max	Scale
kd	Yes	0.001641	1096	Log
mw	Yes	5.744E-5	99	Norm
hb	Yes	1E-6	0.07	Norm
bw	Yes	0.0002634	3345	Log
Fs	No	1	1	Norm

Calibration parameter settings for GUTS-RED-IT:

Parameter	Fit	Min	Max	Scale
kd	Yes	0.001641	73.82	Log
mw	Yes	5.744E-5	70.88	Log
hb	Yes	1E-6	0.07	Norm
bw	No	Inf	Inf	Norm
Fs	Yes	1.05	20	Log

Calibration results

Fitted parameters for GUTS-RED-SD:

Best fit parameter values and their 95% CI

kd: 52.88 (17.18 - 147)

mw: 9.172 (6.935 - 20.09)

hb: 0.0124 (0.004674 - 0.0245)

bw: 0.3047 (0.2075 - 0.7271)

* edge of 95% parameter CI has run into a boundary

(this may also affect CIs of other parameters)

Goodness of fit for calibration data (GUTS-RED-SD):

Model efficiency (NSE, r-square): 0.9843

Normalised root-means-square error (NRMSE): 4.512 %

Minus log-likelihood (MLL): 121.32

AIC: 250.64

Survival probability prediction error (SPPE) for each treatment:

Data set	Treatment	Value
1	Control	2.337 %
1	T1	4.837 %
1	T2	4.837 %
1	T3	1.389 %
1	T4	-21.4 %
1	T5	-8.41E-5 %
1	T6	-3.03E-30 %

GUTS-RED-SD results table for LC_{x,t} [[C]], with 95% CI:

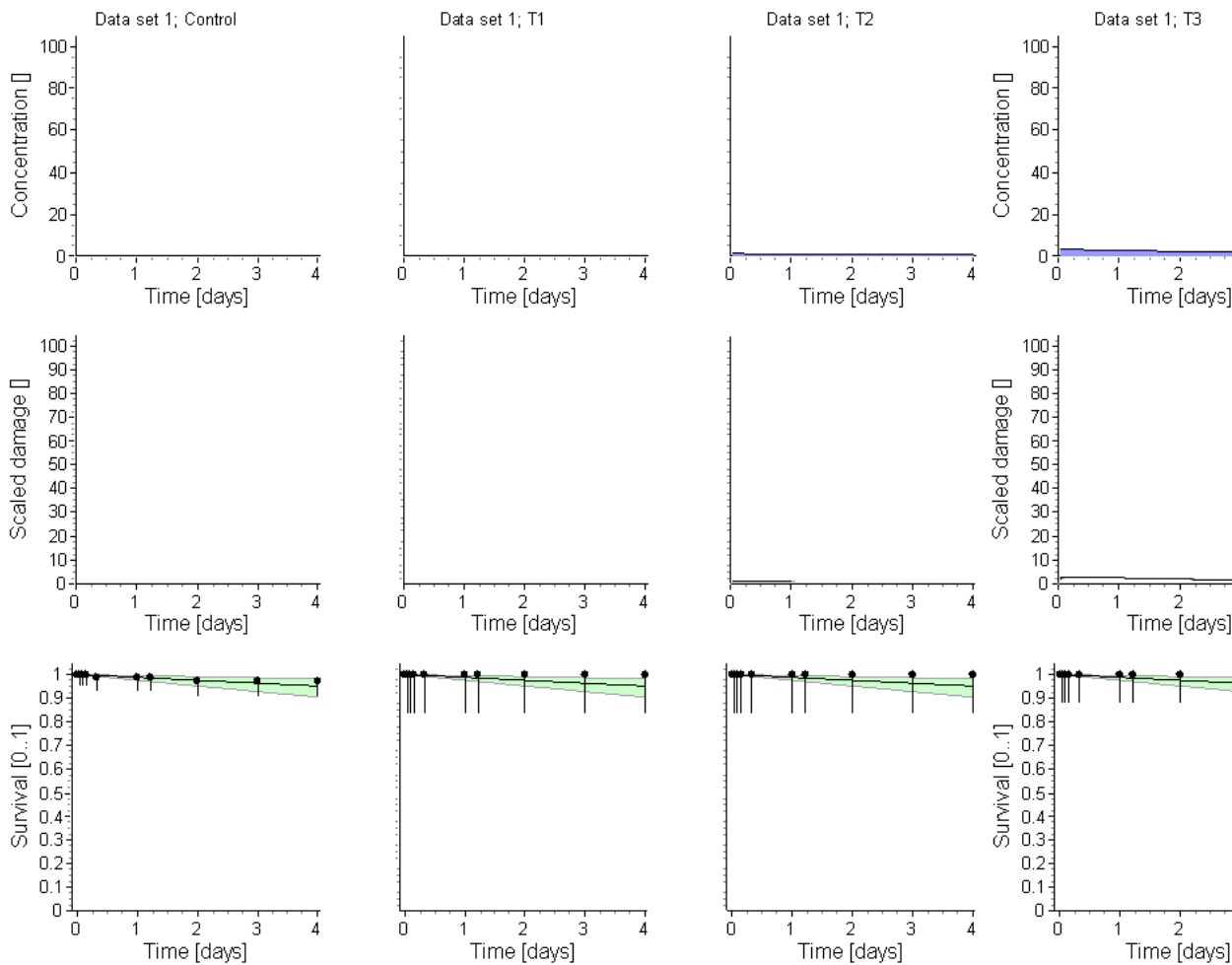
Time [d]	LC50	LC20	LC10
1	11.56 (9.68 - 21.4)	9.956 (7.914 - 20.55)	9.547 (7.417 - 20.33)
2	10.34 (8.377 - 20.74)	9.553 (7.421 - 20.33)	9.353 (7.17 - 20.22)
3	9.947 (7.897 - 20.53)	9.423 (7.259 - 20.26)	9.291 (7.091 - 20.19)

4	9.751 (7.659 - 20.42)	9.359 (7.178 - 20.22)	9.26 (7.053 - 20.17)
7	9.5 (7.355 - 20.29)	9.278 (7.075 - 20.18)	9.222 (7.004 - 20.15)
10	9.401 (7.232 - 20.24)	9.246 (7.034 - 20.16)	9.207 (6.984 - 20.14)
14	9.335 (7.148 - 20.21)	9.224 (7.007 - 20.15)	9.197 (6.971 - 20.14)
21	9.28 (7.078 - 20.18)	9.207 (6.984 - 20.14)	9.188 (6.961 - 20.14)
27	9.256 (7.047 - 20.17)	9.199 (6.974 - 20.14)	9.185 (6.956 - 20.13)
28	9.253 (7.043 - 20.17)	9.198 (6.973 - 20.14)	9.184 (6.955 - 20.13)
42	9.226 (7.009 - 20.15)	9.189 (6.962 - 20.14)	9.18 (6.95 - 20.13)
50	9.217 (6.998 - 20.15)	9.186 (6.958 - 20.13)	9.179 (6.949 - 20.13)
100	9.194 (6.969 - 20.14)	9.179 (6.949 - 20.13)	9.175 (6.944 - 20.13)
182	9.184 (6.956 - 20.13)	9.176 (6.945 - 20.13)	9.174 (6.942 - 20.13)

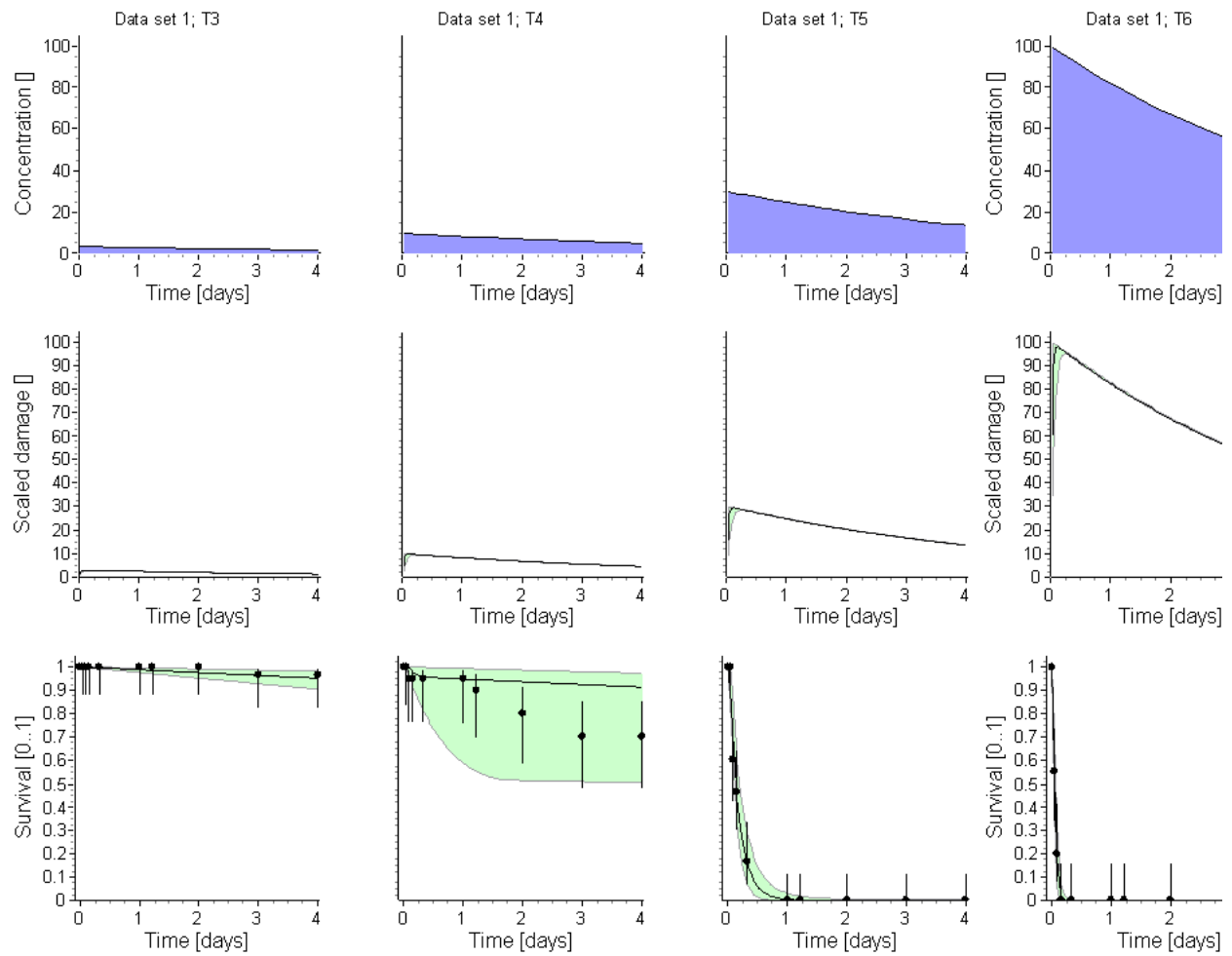
Plots for GUTS-RED-SD calibration:

Parameter space plot for the calibration of GUTS-RED-SD:

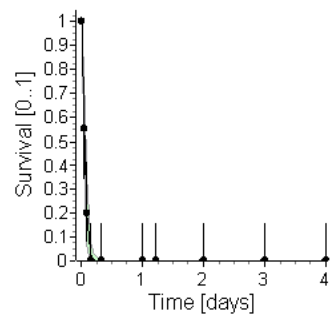
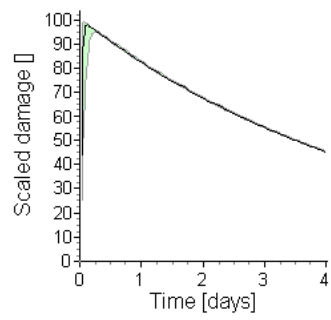
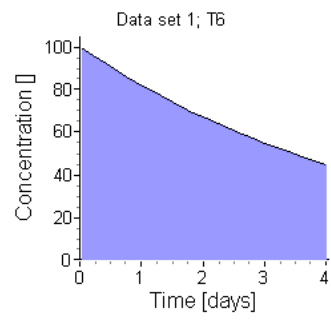
Exposure, damage and survival plots for the calibration of GUTS-RED-SD:



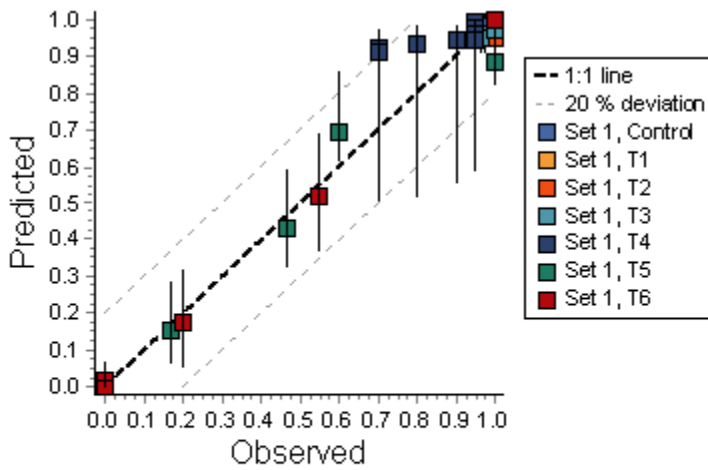
... continued plot:



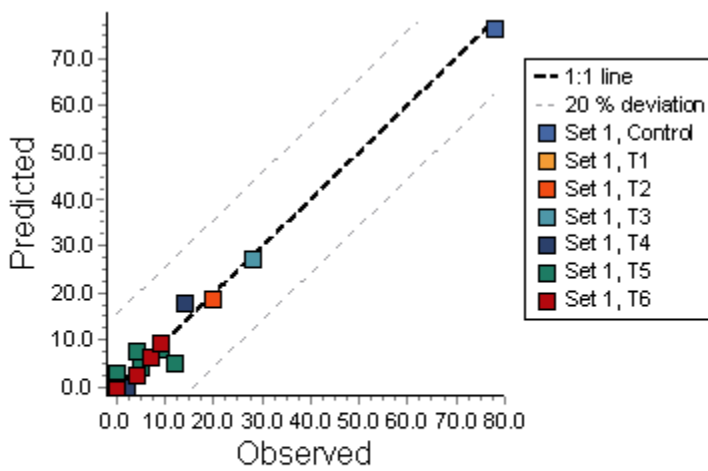
... continued plot:



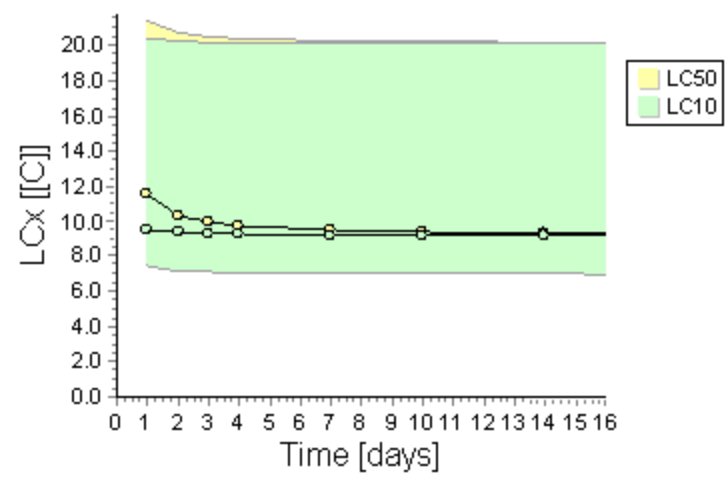
Observed vs. Predicted survival plot for the calibration of GUTS-RED-SD:



Observed vs. Predicted deaths plot for the calibration of GUTS-RED-SD:



LCx versus time with confidence intervals (plotted for 16 days, GUTS-RED-SD):



Fitted parameters for GUTS-RED-IT:

Best fit parameter values and their 95% CI

kd: 3.733 (1.924 - 5.372)

mw: 13.53 (8.64 - 17.82)

hb: 0.01246 (0.005694 - 0.02318)

Fs: 3.359 (2.582 - 4.799)

* edge of 95% parameter CI has run into a boundary

(this may also affect CIs of other parameters)

Goodness of fit for calibration data (GUTS-RED-IT):

Model efficiency (NSE, r-square): 0.9778

Normalised root-means-square error (NRMSE): 6.485 %

Minus log-likelihood (MLL): 130.82

AIC: 269.63

Survival probability prediction error (SPPE) for each treatment:

Data set	Treatment	Value
1	Control	2.36 %
1	T1	4.861 %
1	T2	4.884 %
1	T3	2.04 %
1	T4	-6.56 %
1	T5	-12.3 %
1	T6	-0.37 %

GUTS-RED-IT results table for LC_{x,t} [[C]], with 95% CI:

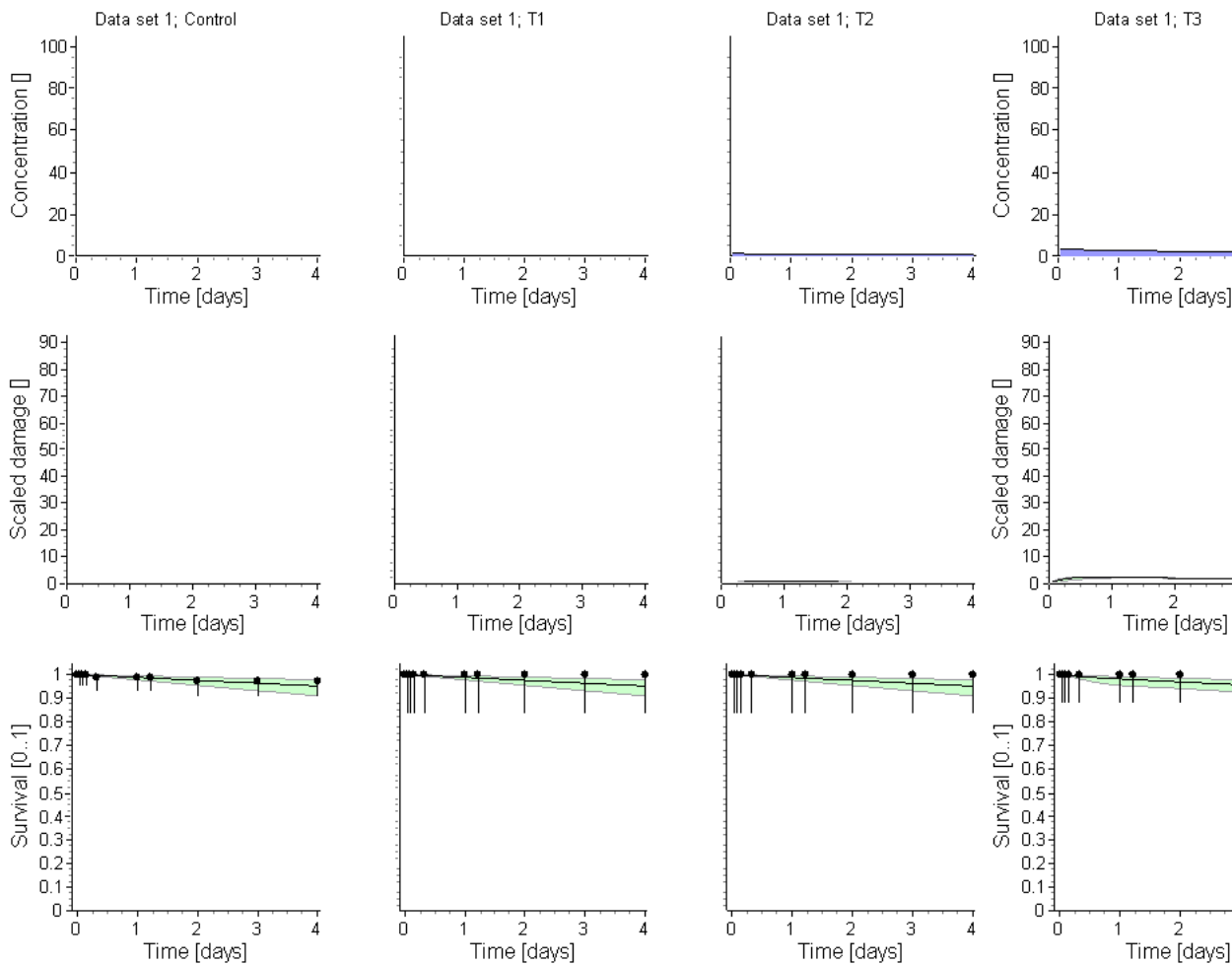
Time [d]	LC50	LC20	LC10
1	13.86 (9.591 - 18.15)	8.764 (5.651 - 11.77)	6.702 (4.147 - 9.317)
2	13.54 (8.138 - 18.03)	8.559 (4.795 - 11.68)	6.546 (3.519 - 9.248)
3	13.53 (7.924 - 18.03)	8.554 (4.669 - 11.68)	6.542 (3.426 - 9.248)
4	13.53 (7.887 - 18.03)	8.554 (4.647 - 11.68)	6.542 (3.41 - 9.248)
7	13.53 (7.879 - 18.03)	8.554 (4.642 - 11.68)	6.542 (3.407 - 9.248)

10	13.53 (7.879 - 18.03)	8.554 (4.642 - 11.68)	6.542 (3.407 - 9.248)
14	13.53 (7.879 - 18.03)	8.554 (4.642 - 11.68)	6.542 (3.407 - 9.248)
21	13.53 (7.879 - 18.03)	8.554 (4.642 - 11.68)	6.542 (3.407 - 9.248)
27	13.53 (7.879 - 18.03)	8.554 (4.642 - 11.68)	6.542 (3.407 - 9.248)
28	13.53 (7.879 - 18.03)	8.554 (4.642 - 11.68)	6.542 (3.407 - 9.248)
42	13.53 (7.879 - 18.03)	8.554 (4.642 - 11.68)	6.542 (3.407 - 9.248)
50	13.53 (7.879 - 18.03)	8.554 (4.642 - 11.68)	6.542 (3.407 - 9.248)
100	13.53 (7.879 - 18.03)	8.554 (4.642 - 11.68)	6.542 (3.407 - 9.248)
182	13.53 (7.879 - 18.03)	8.554 (4.642 - 11.68)	6.542 (3.407 - 9.248)

Plots for GUTS-RED-IT calibration:

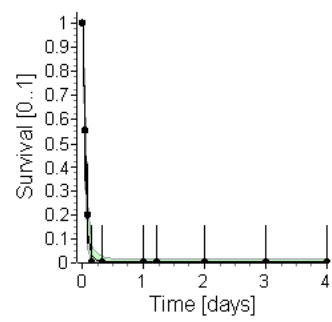
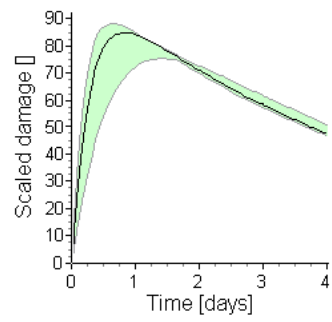
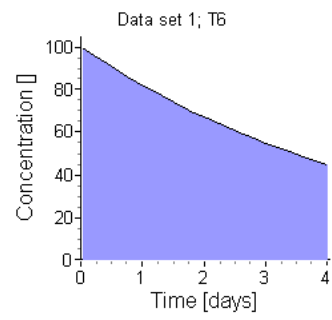
Parameter space plot for the calibration of GUTS-RED-IT:

Exposure, damage and survival plots for the calibration of GUTS-RED-IT:

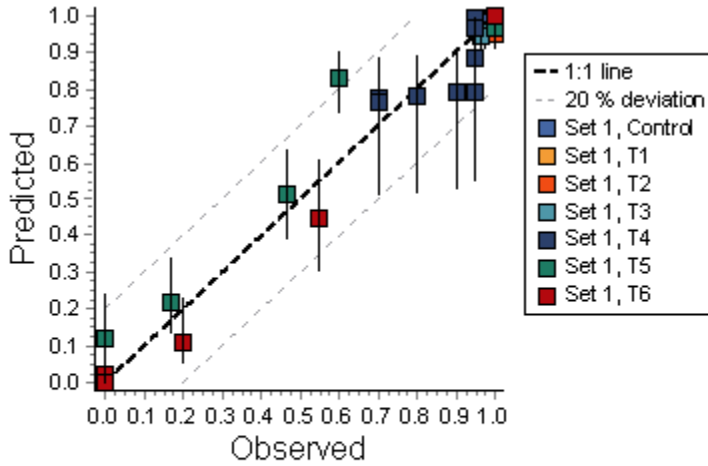


... continued plot:

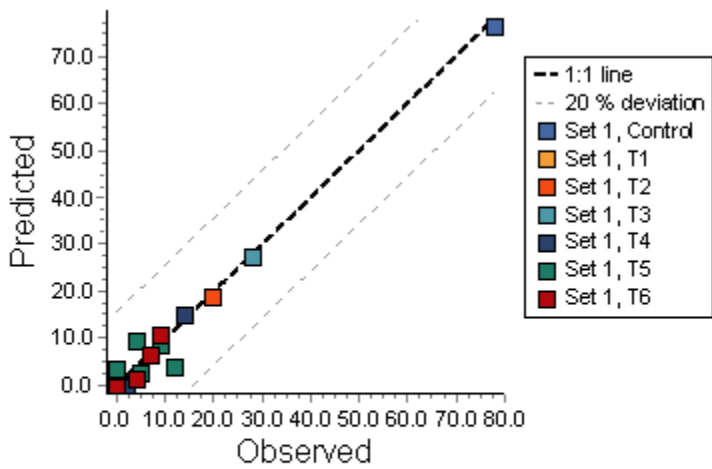
... continued plot:



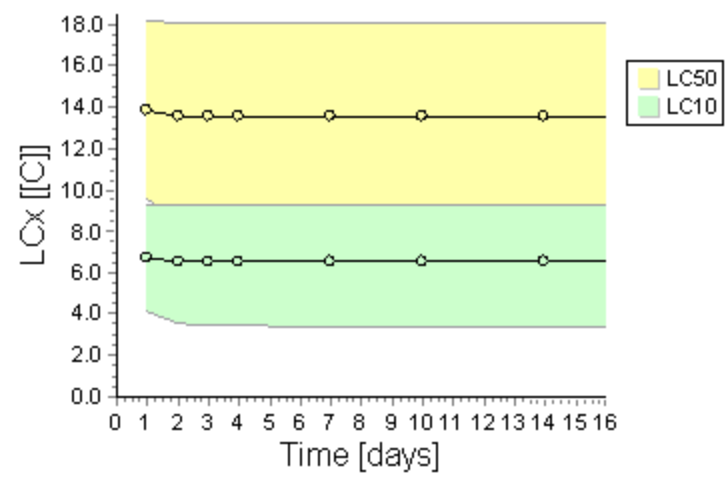
Observed vs. Predicted survival plot for the calibration of GUTS-RED-IT:



Observed vs. Predicted deaths plot for the calibration of GUTS-RED-IT:



LCx versus time with confidence intervals (plotted for 16 days, GUTS-RED-IT):



Validation

No validation performed!

Predictions

No predictions performed!